Closed-loop cell therapy

A virtual CAR-T perfusion bioreactor and its digital shadow, built on Egan et al. (2025) and explained for a CHO engineer

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Repository `closed-loop-cell-therapy`, version 0.2.0

The project in four sentences

A patient's own T cells are expanded for five to six days in a small closed bioreactor, the Aglaris FACER, and the cells themselves are the drug. The cell concentration cannot be measured during the run without disturbing the cells, yet it decides when the batch reaches its dose and can be harvested. Egan et al. (2025) built a *digital shadow*: a mechanistic model fitted in real time to the signals the bioreactor already measures (perfusion rate, glucose, lactate), which forecasts the cell concentration. This repository rebuilds their model and their six published runs as a virtual bioreactor, and rebuilds the shadow on top of it.

Source. Egan JR, Marí-Buyé N, Benítez-Cano EV, et al. A digital shadow of CAR T cell expansion in a perfusion bioreactor: informing optimal harvest times for autologous cell therapy. *Biotechnology Progress* 2025;41(6):e70045, doi:10.1002/btpr.70045. Open access (CC BY 4.0). Equation numbers below refer to the paper (1–18) and its supporting information (S1–S80).

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1 Cell therapy, translated from CHO

In a CHO process the cells are a factory: they secrete an antibody and are discarded at harvest. In autologous CAR-T the *cells are the product*. T cells are collected from the patient, activated so that they divide, given a gene for a chimeric antigen receptor (CAR) that makes them recognise the tumour, **expanded**, then harvested, formulated and infused back into the same patient. The repository models the expansion.

	CHO (mAb)	CAR-T in the FACER
Product	secreted protein	the cells
Batch	one cell line, many patients	one patient, one batch, 134 mL chamber
Starting material	characterised cell bank	the patient's blood; varies between patients
Quantity attribute	titre	dose: number of cells
Quality attribute	glycosylation and other variants	cell phenotype; worsens with long expansion
Vessel	stirred tank, cells in suspension	static chamber, cells sedimented on a gas-permeable membrane
Perfusion	ATF/TFF with cell bleed	medium in and out above the settled cells, no bleed
Cell count during the run	routine sample or capacitance probe	not possible without resuspending the cells

The harvest decision. Longer expansion gives more cells but is associated with progressive differentiation and less effective cells, and with higher cost and longer waits for patients. Too short an expansion may not reach the dose, especially from heavily pretreated patients. The best harvest time is therefore patient specific, and it needs the cell concentration, which is exactly what cannot be measured during the run.

Metabolism is familiar: activated T cells are strongly glycolytic. Across the five training runs the yield of lactate from glucose averages 1.99 mol/mol, close to the 2 of aerobic glycolysis (Table 2 of the paper).

2 The FACER process

The FACER cartridge is a closed, single-use, automated system with two culture chambers (13.4 mL on 10 cm^2, and 134 mL on 100 cm^2). Cells grow under static conditions on a gas-permeable membrane that separates the medium from a controlled gas zone. Small starting numbers are seeded in the small chamber and passaged to the large one; all modelling here is of the large chamber.

1. **Seeding** at t_I with concentration X_I, counted offline.
2. **Sedimentation**, t_I to t_{sed}: no perfusion; the cells settle on the membrane.
3. **Perfusion**, t_{sed} to t_F: fresh medium (glucose S_I) enters and spent medium leaves at the same rate D, so the volume is constant. A PID controller sets D to hold glucose at a set point S_{sp}. Cells stay sedimented and none leave.
4. **End**: cells are resuspended and counted, giving X_F.

Online, single-use sensors read glucose and lactate every 10 s (enzymatic), plus pH and dissolved oxygen. Bubbles passing the sensors cause false drops in glucose and lactate, so the raw signals go through a 2nd-order low-pass Butterworth filter (cut-off 0.001 Hz) before the controller sees them. Some artefacts survive: downward glucose spikes, then upward perfusion spikes.

3 The model

Time t is in days, cells X in 10^6 cells/mL, glucose S and lactate P in mmol/L. With Y_{xs} in cells/pmol and Y_{px} in pmol/cell, the terms $\mu X/Y_{xs}$ and $Y_{px}\mu X$ come out directly in mmol/L/day. H is the Heaviside step.

3.1 Growth: the Richards model, early on

Growth is limited not by glucose or lactate (kept in comfortable ranges by perfusion) but by cells competing for space. Early in the expansion, far below the carrying capacity, the Richards model reduces to a two-parameter form, with a lag:

$$X(t) = \frac{X_I}{\left(1 - H(t - t_{\text{lag}})\,\mu_{\max}\,\nu\,(t - t_{\text{lag}})\right)^{1/\nu}}, \tag{1}$$

$$\mu(t) = \frac{H(t - t_{\text{lag}})\,\mu_{\max}}{1 - H(t - t_{\text{lag}})\,\mu_{\max}\,\nu\,(t - t_{\text{lag}})}, \qquad \nu < 0. \tag{2}$$

These are eqs 10 and 11 (S60, S61). $\mu_{\max}$ is the growth rate at the end of the lag and ν sets how fast it decays. Special cases (S46–S50): $\nu = -1$ gives linear growth, $\nu = -\frac{1}{2}$ quadratic growth, and $\nu \to 0$ exponential growth (figure 1).

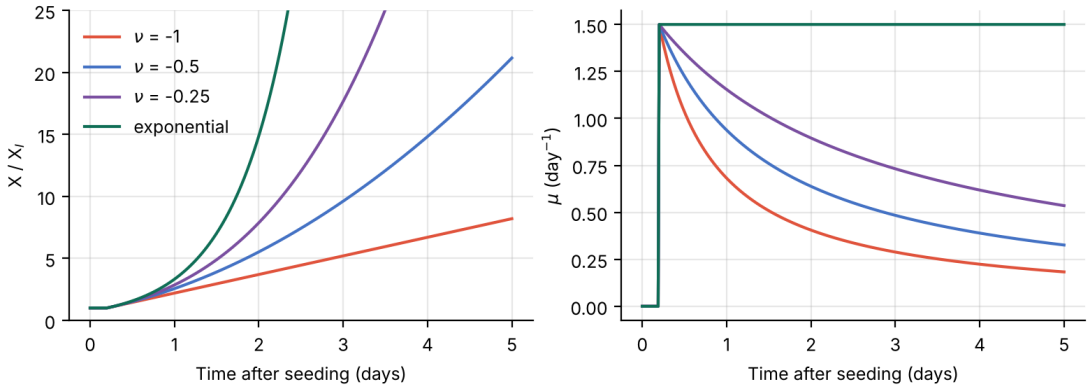


Figure 1: The growth model for $\mu_{\max} = 1.5$/d and a lag of 0.2 day. The more negative ν, the faster the growth rate falls.

Two closed forms are used throughout. If both counts are known, $\mu_{\max}$ follows from the others (eq. 12), and the time to reach a threshold X_T, for example a dose, is eq. 13:

$$\mu_{\max} = \frac{1 - (X_I/X_F)^{\nu}}{\nu\,(t_F - t_{\text{lag}})}, \qquad t_T = t_{\text{lag}} + \frac{1 - (X_I/X_T)^{\nu}}{\mu_{\max}\,\nu}. \tag{3}$$

3.2 Balances

$$\frac{\mathrm{d}S}{\mathrm{d}t} = H(t - t_{\text{sed}})\,D\,(S_I - S) - \frac{\mu X}{Y_{xs}}, \tag{4a}$$

$$\frac{\mathrm{d}P}{\mathrm{d}t} = Y_{px}\,\mu X - H(t - t_{\text{sed}})\,D\,P, \tag{4b}$$

$$\frac{\mathrm{d}D}{\mathrm{d}t} = H(t - t_{\text{sed}})\left[-K_p\,\frac{\mathrm{d}S}{\mathrm{d}t} + K_i\,(S_{\text{sp}} - S)\right], \qquad D \geq D_{\min}, \tag{4c}$$

$$\frac{\mathrm{d}C'}{\mathrm{d}t} = \left(H(t - t_{\text{sed}})\,D + k_L a\right)(1 - C') - \frac{\mu X}{Y'_{xc}}. \tag{4d}$$

Equation (4c) is the velocity form of a PI controller (eq. 2, S18); the derivative gain is negligible. C' is dissolved oxygen as a fraction of saturation (eq. 8), with Y'_{xc} the scaled yield of cells on oxygen. The lactate yield from glucose is $Y_{ps} = Y_{px}Y_{xs}$ (eq. 5).

Lactate settles by itself. Once glucose sits at its set point, $\mathrm{d}S/\mathrm{d}t \approx 0$ and the glucose balance gives $D\,(S_I - S_{\mathrm{sp}}) = \mu X/Y_{xs}$. Substituting into the lactate balance, $\mathrm{d}P/\mathrm{d}t = D\left(Y_{ps}(S_I - S_{\mathrm{sp}}) - P\right)$, so

$$P \to P_{ss} = Y_{ps}\,(S_I - S_{\mathrm{sp}}) \qquad \text{(eq. 18)}, \tag{5}$$

whatever the growth. And the perfusion rate tracks growth: with $\nu = -1$ (linear growth) it is roughly constant, with $\nu = -\frac{1}{2}$ it rises linearly (eq. 15).

Numerics. The paper found that `solve_ivp` let the perfusion rate go negative, so the ODEs are discretised by forward Euler with a 10 s step and each state is clipped at its bound every step (S22–S24): $S, P \geq 0$, $D \geq D_{\min}$. $X(t)$ and $\mu(t)$ are analytical. `process/facer.py` implements exactly this.

3.3 The six published runs

Run	Cells	t_I	t_{sed}	t_F	S_{sp}	X_I	X_F	Y_{xs}	Y_{px}	$t_{\mathrm{lag}} - t_I$	ν	$\mu_{\max}$	k_La
1	donor 1, A	0.11	0.34	5.05	20	1.49	19.0	0.231	7.93	0.160	−0.922	2.14	3.00
2	donor 2, A	0.14	1.02	6.03	20	0.37	19.0	0.275	6.25	0.172	−0.477	2.02	741
3	donor 3, A	0.11	0.86	6.01	20	1.49	32.5	*0.259*	*7.62*	*0.092*	*−0.537*	*1.36*	*3.00*
4	patient 1, A	2.09	2.84	7.00	20	0.62	10.9	0.272	9.09	0.128	−0.524	1.39	1.73
5	donor 4, A	1.92	2.92	6.95	17	1.13	8.73	0.207	6.83	0.000	−0.000	0.405	42.9
6	donor 5, B	0.08	0.83	4.85	21	1.49	18.2	0.311	7.99	0.000	−0.761	1.58	20.2

Times and concentrations are from Table 1, yields, lag and ν from Table 2, $\mu_{\max}$ from eq. 12, k_La and Y'_{xc} from Table S4. All runs share $S_I = 25$ (26 for run 6), $K_p = 0.215$, $K_i = 5.16$, $K_d = 7.46 \times 10^{-5}$ and $D_{\min} = 0.067/\mathrm{d}$. Media A is TexMACS with IL-7 and IL-15; media B is Advanced TCM with IL-2. Run 4 is the patient's CAR-T cells. *Italics*: run 3 was the paper's independent test run and was never fitted; here it uses the training means, with $\mu_{\max}$ set by eq. 12 so that it ends at the published X_F.

4 The virtual FACER

The model above is what the shadow fits. The *virtual bioreactor* that generates the data (`sim/runner.py`) uses the same balances, but adds what makes real data hard:

- **Sensors**: noisy enzymatic glucose and lactate every 10 s, with bubbles drawn as a Poisson process, then the published Butterworth filter, applied causally.
- **The controller acts on measurements**: the PID (`control/perfusion.py`, the velocity form of S12 with the small derivative term) sees the filtered glucose, never the truth.
- **Offline counts** only at seeding and at the end, with 5% error.
- **Faults**: run 3's pump stop, 2.5 days into perfusion for half a day, during which perfusion stops and the online data are lost. Optional slow random variation of growth.

5 The digital shadow

A *digital twin* exchanges data both ways and can act on the process; a *digital shadow* only receives data and advises a person. The shadow here (`shadow/fit.py`) sees what the platform logs: perfusion rate, filtered glucose and lactate, and the seeding count. It simulates the whole controlled process, PI included, and fits it to the three online signals at once, with equal weights, by least squares.

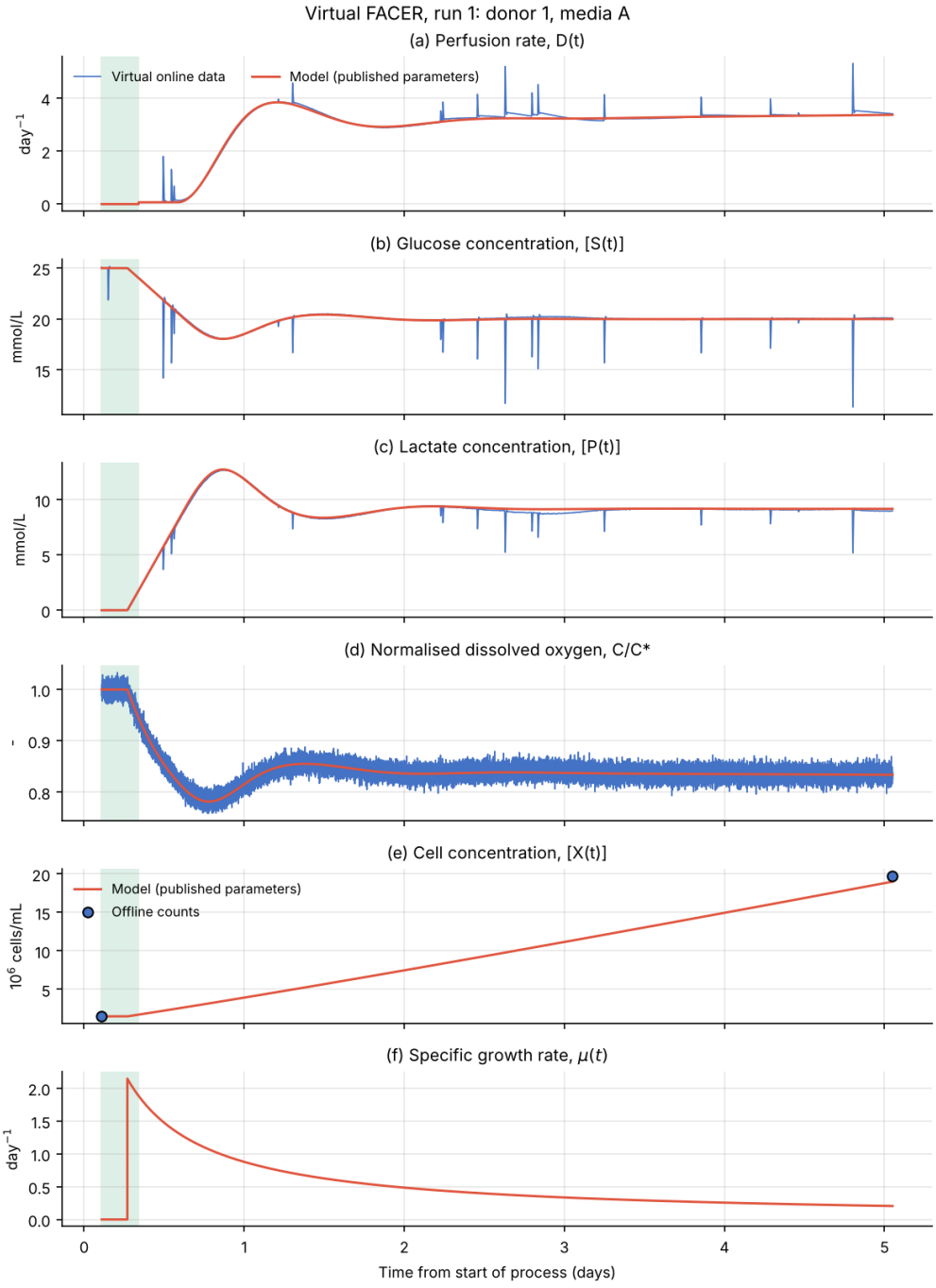


Figure 2: Virtual run 1 (blue) against the model with the published parameters (red), laid out like Figure 3 of the paper. Green band: sedimentation. The controller overshoots once, then holds glucose at 20 mM; lactate settles at P_{ss}; bubble artefacts give glucose dips and perfusion spikes.

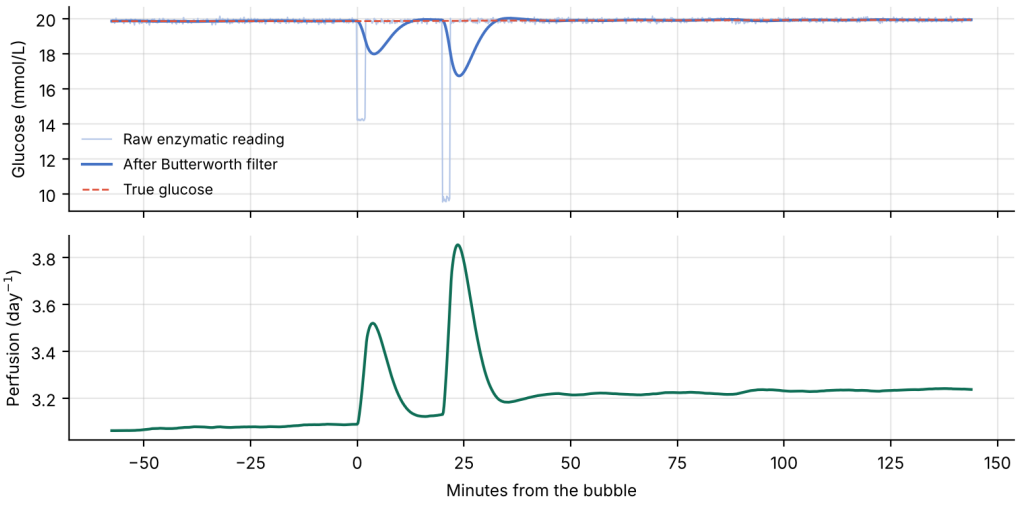


Figure 3: Two bubbles, minute by minute. The filter turns a deep, short drop in the raw reading into a shallow, longer dip; the controller still reacts with a perfusion spike.

Mode	Known	Fitted	Fixed
Training	X_I, X_F, all online data	Y_{xs}, Y_{px}, t_{lag}, ν	$\mu_{\max}$ from eq. 12
Real time	X_I, online data so far	Y_{px}, $\mu_{\max}$, ν	$t_{\text{lag}} - t_I = 0.092$ and $Y_{xs} = 0.259$, the training means

Why fix two parameters in real time. The real-time model is structurally identifiable only if $\mu_{\max}$ or t_{lag} is known (Table S1). Fixing the lag alone still left the fit practically unidentifiable, so the paper also fixed Y_{xs}, the most consistent parameter across runs. The DO equation is left out because $k_L a$ and Y'_{xc} are not practically identifiable from these data (Table S4). A CHO analogy: you can infer cell growth from how hard the controller works to hold glucose, but only if you know how much glucose each new cell costs.

Confidence bands propagate the covariance of $(\mu_{\max}, \nu)$ to $X(t)$ and $\mu(t)$ to first order, using analytical sensitivities (S67–S70), scaled by the reduced χ^2 and a Student t factor (S71, S72). The fit uses scipy's trust-region least squares with bounds ($\nu < 0$, positive yields) on a 60 s grid; the paper used Levenberg–Marquardt via lmfit at 10 s.

5.1 Results

Training fits recover the published parameters to within a few per cent; the residual differences come from the 5% counting error on X_F and the sensor noise. Figure 5 repeats the paper's pseudo-real-time test.

Days of data	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Mean $\lvert\cdot\rvert$
1.5	28.2	12.7	−4.6	0.6	−5.3	−31.9	13.9
2.5	15.8	−1.5	−3.9	−5.5	−0.6	−22.7	8.3
4.0	7.7	−7.6	–	−7.4	15.7	−17.8	11.3

Relative error (%) of the predicted final cell concentration. With 2.5 days of data every run is within 30%, with a mean absolute error of 8% (the paper: 13%, s.d. about 6%).

What the errors say. Runs 1 and 6 stay biased however much data arrives, in opposite directions. Their true Y_{xs} (0.231 and 0.311) differs from the fixed mean (0.259) by about −11% and +20%, and the shadow converts glucose demand into cells through Y_{xs}, so the forecast inherits that bias. The bands do not cover it, because they only carry the uncertainty of the fitted parameters. The paper

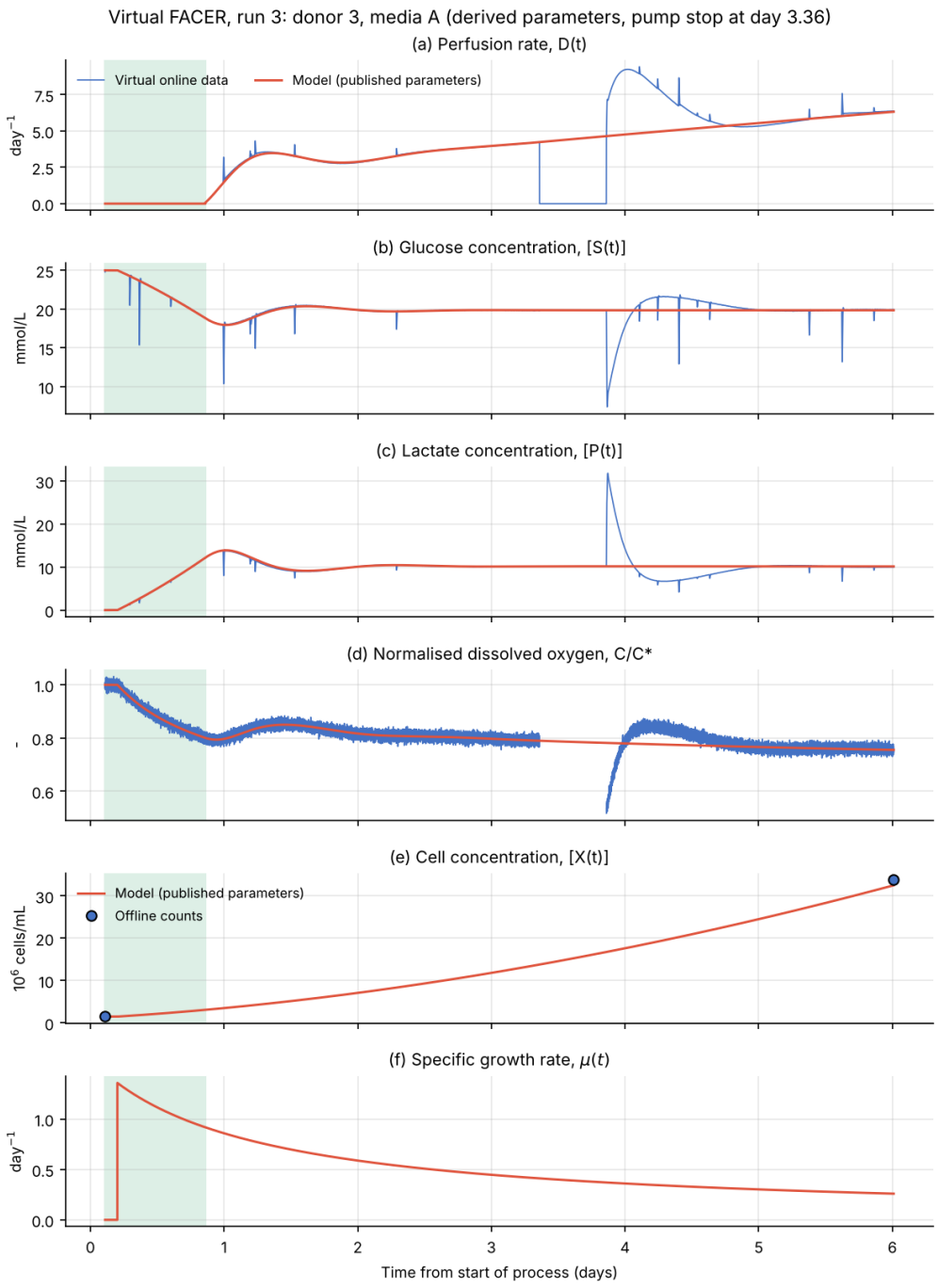


Figure 4: Virtual run 3 with its pump stop. Without perfusion, glucose falls and lactate climbs; when the pump restarts, the controller pushes the perfusion rate well above the model line to recover. The model assumes growth is not affected, which is the paper's assumption too.

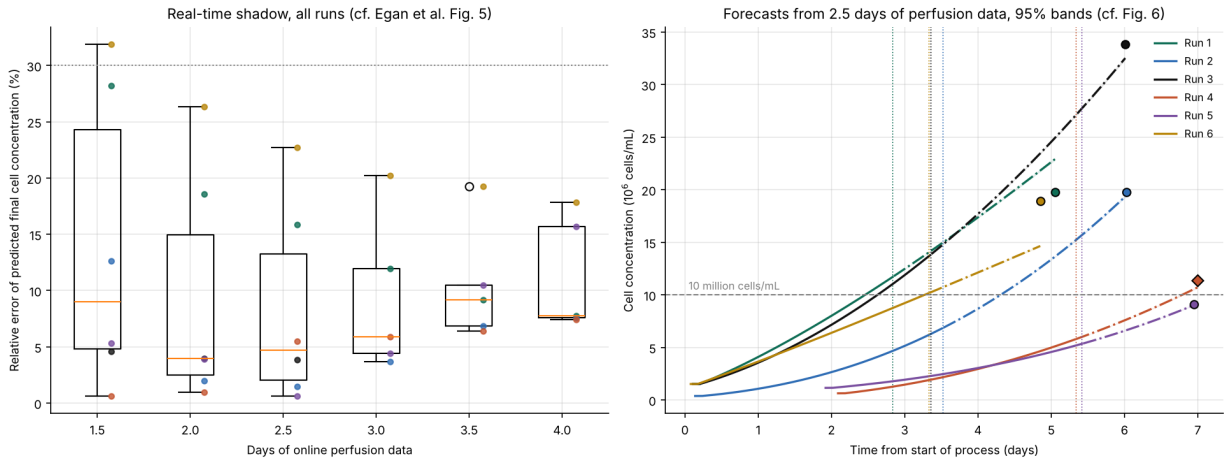


Figure 5: Left: absolute relative error of the predicted final concentration against days of perfusion data (cf. Figure 5 of the paper). Right: forecasts from 2.5 days of data, with the offline final counts (cf. Figure 6). The 95% bands are almost invisible.

reached the same conclusion and proposed the fix: sample Y_{xs} and t_{lag} from their distribution across runs and refit for each sample. That is an ensemble method, which is where an EnKF version of the shadow comes in (day 6).

6 Checks

`pytest` runs 31 tests, including:

- the growth special cases (linear, quadratic, exponential) and $\mu = \mathrm{d}\ln X/\mathrm{d}t$;
- $\mu_{\max}$ and the expansion periods to 10^7 cells/mL against Table 2, for all five training runs;
- eqs 12 and 13 as inverses, and the analytical sensitivities against finite differences;
- every run reaching its published X_F, glucose at set point and lactate at P_{ss};
- no perfusion during sedimentation, $D \geq D_{\min}$ after it, and the PID's behaviour;
- the filter's unit gain and lag, bubbles pulling readings down, glucose held despite them;
- run 3 losing perfusion and data during its stop;
- the training fit recovering run 2's parameters, the 2.5-day forecast within 30%, bands narrowing with more data, and the harvest time matching the forecast.

7 Limitations and next steps

- **Synthetic data.** The paper's online data are proprietary. The virtual runs reproduce the published parameters and the shape of the published traces, not the traces themselves.
- **Inherited assumptions**: a well-mixed chamber although the cells are sedimented; no nutrient limitation, death or maintenance; constant yields; all cells alike (no CD4/CD8 or phenotype segregation).
- **Placeholders**: sensor noise and bubble statistics; run 3's parameters.
- **Next**: expose the virtual FACER over OPC UA (day 4); an EnKF version of the shadow that samples Y_{xs} and t_{lag} (day 6); harvest recommendations with an operator in the loop (day 8); and using the shadow to tune the PID, the paper's suggested use for run 6's oscillations (day 10).

8 Code map

`process/growth.py`	growth model, eqs 10–13, S60–S70
`process/parameters.py`	the six runs with their sources; training means
`process/facer.py`	balances and forward Euler, S19–S24; the model the shadow fits
`control/perfusion.py`	glucose PID on the perfusion rate
`control/pid.py`	generic PI with anti-windup, the example in the control refresher notes
`sensors/online.py`	enzymatic sensors, bubbles, Butterworth filter, DO, offline counts
`sim/runner.py`	the virtual FACER, faults, history and CSV export
`shadow/fit.py`	training and real-time fits, bands, harvest time
`scripts/simulate_run.py`	figures 2 and 4
`scripts/realtime_test.py`	figure 5 and the error table
`docs/technical_documentation/make_figures.py`	all figures in this document